

Cluster Robust Variance Estimation

Edoardo Di Cosimo

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1 Introduction

In constructing the confidence intervals of the OLS estimator $\hat{\beta}_{\text{ols}}$, it is the asymptotic distribution of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i$$

that determines the estimator's asymptotic distribution. Assuming that $E[X_i u_i] = 0$, the Lindberg-Levy CLT apply:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N} \sum_i \text{Var}(X_i u_i)\right)$$

But this hold true only if the individual observations are all independent, in fact summing 2 random variables X_1 and X_2 we get another random variable (call it Z) with expected value equal to $E[X_1] + E[X_2]$ and variance equal to $\text{Var}(X_1) + \text{Var}(X_2) + 2 \cdot \text{cov}(X_1, X_2)$. Summing N random variables $(X_1, \dots, X_n)$, assume $E[X_i] = 0$, we get a new random variable $Z_N = \sum_{i=1}^N X_i$:

$$Z_N \sim \left(0, \sum_{i=1}^N \text{Var}(X_i) + 2 \sum_{i \neq j}^N \sum_{j=1}^N \text{cov}(X_i, X_j)\right)$$

if all the X_i s are **independent** all the joint moments (and so also the covariances) are equal to zero and, dividing by $\sqrt{N}$:

$$\frac{1}{\sqrt{N}}(Z_N) \sim \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i)\right)$$

So the variance of Z_N scaled by $\frac{1}{\sqrt{N}}$ is the average variance of all the X_i s. But here comes the problem, if our data are clustered we do not have that all the observation are independent, but rather we have that the observation of all the individuals inside a cluster are correlated, so we would have that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i) + \frac{2}{N} \sum_{i \neq j}^N \sum_{j=1}^N \text{cov} (X_i u_i, X_j u_j) \right)$$

The variance estimator that is most commonly used, the White variance estimator assumes that all the covariances in the expression above are 0, and so estimates only $\frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i)$, so it would not consistently estimate the asymptotic variance.

1.1 Why we estimate $\text{Var} (\hat{\beta})$

To follow what comes next, it is essential to clarify **what we are actually estimating** when we estimate the variance of

$\hat{\beta}_{\text{ols}}$. After all, $\hat{\beta}_{\text{ols}}$ is just a number—so what does it mean to speak of its variance?

Is it a random variable? Does it have a distribution? Yes. The distribution we refer to is the distribution of $\hat{\beta}_{\text{ols}}$ **across repeated samples**, i.e., the values it would take if we changed the individuals drawn from the population. Since $\hat{\beta}_{\text{ols}}$ is a function of random variables (X_i and u_i), if we knew their full population distributions, we would know the exact value of β . But because we only observe a sample, our estimate varies from sample to sample. In this sense, $\hat{\beta}_{\text{ols}}$ is a random variable and therefore has a distribution.

1.2 What are clustered data?

Let's take a step back and ask ourselves what does it mean that our data are clustered? A. C. Cameron and D. L. Miller [1] define a cluster as a subgroup in the population such that individuals inside it has errors that are correlated. According to A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge [2] what matters is the correlation of $X_i u_i$. This difference arises due to the fact that in A. C. Cameron and D. L. Miller [1] regressors are considered non stochastic. In essence, the defining feature of a cluster is that individuals within it exhibit some form of **unobserved** correlation.

Suppose our sample contains N individuals, each with an observed trait X . We can represent these as $X_{ig} = (X_{11}, \dots, X_{N_1 1}, X_{12}, \dots, X_{N_2 2}, \dots, X_{N_G G})$, where i indexes individuals within a cluster and g indexes the cluster.

We can think of all these X_{ig} as being drawn from the same population distribution, so they are **identically distributed**. Let us assume this distribution has mean 0.

However, individuals belonging to the same cluster are **not independent**. Their joint distribution is not simply the product of their marginals, their covariances may not be zero.

Saying that observations within a cluster are correlated means that knowing the value of one

individual gives information about another individual in the same cluster. If individuals i and j belong to the same cluster g , then:

$$P(X_i > E[X] \mid X_j > E[X], i, j \in g) > P(X_i > E[X])$$

and similarly:

$$P(X_i < E[X] \mid X_j < E[X], i, j \in g) > P(X_i < E[X]).$$

outcomes inside the same cluster tend to move together relative to the overall mean—they are positively associated. If we focus on a single cluster g , we can view its observations as:

$$X_g = \{X_i \mid \text{individual } i \in \text{cluster } g\}$$

Each cluster is a subpopulation with its own distribution, and the overall population is the mixture of all G cluster distributions.

1.3 General setup

In this section, we lay out the general setup that will guide the rest of the thesis. We assume that there are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster.

$$y_{ig} = \alpha + \beta_g X_{ig} + u_{ig}$$

We model u_{ig} has having a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level. In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_g(u_{ig}) = c_g$$

But as long as we have an intercept that varies for each cluster we can assume w.l.o.g that $\mathbb{E}[u_{ig}] = 0$.

We will call β_{APE} the average between cluster of the various β_g .

If we estimate β separately across G clusters, we obtain a vector of cluster-specific estimates $\{\hat{\beta}_g\}_{g=1}^G$.

$$\hat{\beta}_g = \beta_g + \left(\sum_{i \in g} X_i^2 \right)^{-1} \left(\sum_{i \in g} X_i u_i \right)$$

And:

$$\hat{\beta}_g \stackrel{a}{\sim} \mathcal{N} \left(\beta_g, \frac{1}{N_g} \left(\sum_{i \in g} X_i^2 \right)^{-1} B \left(\sum_{i \in g} X_i^2 \right)^{-1} \right)$$

where B_g is:

$$B_g = \text{Var} (X_{ig}u_{ig})$$

If for each cluster we could consistently estimate its asymptotic variance, we could take the average of these cluster-level estimates to obtain a consistent estimate of the asymptotic variance of the overall $\hat{\beta}_{\text{pols}}$, since a linear combination of normal random variables is also normal. But first we have to ask ourselves whether each of the estimated parameters is consistent.

2 Assumption we make

We must begin by defining precisely the assumptions that we are willing to make about the population.

The first choice is whether we treat the number of clusters as fixed and fully observed in every sample—for example, U.S. states ($G = 50$) when estimating the effect of a federal policy. Alternatively, we may assume an increasing number of clusters, where the specific clusters included can vary from sample to sample. In what follows, we discuss both cases.

Fixed amount of clusters

We can think of the data generating process (DGP) as having 2 stages:

- in the first stage all the population parameters are drawn. This includes the vector of length G with all the β_g , and the cluster specific distributions for X_g and u_g . This stage only happens once, all the parameters from this stage are fixed.
- The second stage is a draw from the cluster distributions of (X, β, u) of the individual observations that will make our sample. This stage re-happens every time we take a sample from the population, this is the source of randomness.

Infinite amount of clusters

If the amount of clusters that we possibly have is not finite the first sampling stage reoccurs every time we sample. So there are two sources of randomness:

1. which clusters we draw from the possibly infinite population of clusters and,
2. which individuals we draw from those clusters.

2.1 Endogeneity

Then we must decide whether the sample we are working with presents cluster-level endogeneity, i.e. if $\mathbb{E}[X_i u_i \mid i \in g] \neq 0$. In order for $\hat{\beta}_{\text{ols}}$ to be consistent we need exogeneity, we need the regressors to be uncorrelated with the errors, but it may be that this requirement is met at the population level but not in all clusters, it may be that some clusters present correlation between the cluster specific portion of the errors and the regressors (A. C. Cameron, D. L. Miller, and others [3]), and that this correlation averages out across clusters. This would result in a biased estimation of the $G \hat{\beta}_g$, but in a consistent estimation of the average partial effect.

3 Simulating a dataset

We will simulate data in two ways, in the first one we will make so that the expected value of the score is 0 in all clusters, in the second one the expected value of the score will be 0 only at the population level.

First we draw a random vector β_g from a normal distribution with mean β_{APE} and variance σ_β^2 .

```
include("../src/ClusterSimulation.jl")
G = 50 #Number of clusters
Ng = 100 # Number of individuals per cluster

β = rand(Normal(2,4),G)
```

```
50-element Vector{Float64}:
-1.1136689049261572
-2.5887463353250793
-3.5355236251089863
 3.0218633583906906
-0.6195225964387663
 4.3063601947293115
-0.2609448691964169
-0.13059255371766065
-0.3930045979778143
-1.4806951871603755
-3.849905964060212
 3.581254107260582
-3.91541667282402
 ⋮
 3.656387818243681
 2.4166475520672823
-1.4650980389601473
 1.725503730688546
 3.577109201704825
 5.738079037286744
 0.6846812547735872
10.353983510168273
 7.942556319757224
 1.8898351432373293
 2.1066236050641045
-2.9149632077915744
```

We then simulate our dataset in two steps, according to the following model:

$$y_{ig} = \beta_g x_{ig} + (c_g + v_{ig})$$

In the first step we simulate the population parameters, and in the second we draw from them.

3.1 Description of the Two Steps

```
pop = hePopulation(G, β, randn(G))
```

Population with 50 clusters and with heterogeneous effect

This line creates a heterogeneous-effects population.

- **G** is the number of clusters.
- **β** is the vector of betas.
- **randn(G)** produces the vector $c_g \sim \mathcal{N}(0, 1)$.

The constructor returns a population struct that stores the cluster-level parameters used to simulate data.

```
sam = sample(pop, Ng)
```

Cluster sample struct

This line generates a sample from the population.

- **Ng** is the number of observations per cluster.

The function draws data for all **G** clusters, each containing **Ng** observations, and returns a sample struct built according to the population's parameters.

$\hat{\beta}_g$ in each cluster and then broadcast it to all clusters in our sample, getting G estimates.

4 Het

There are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster.

$$y_{ig} = \alpha + \beta_g X_{ig} + u_{ig} \quad (1)$$

assume u_{ig} has a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level.

In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_g(u_{ig}) = c_g$$

But as long as we have an intercept we can assume w.l.o.g that $\mathbb{E}[u_{ig}] = 0$. If I estimate $\hat{\beta}_{ols}$ inside cluster g , I get:

$$(\hat{\beta}_g - \beta_g) = \left(\sum_{i \in g} X_i^2 \right)^{-1} \left(\sum_{i \in g} X_i u_i \right)$$

And:

$$\hat{\beta}_g \stackrel{a}{\sim} \mathcal{N}\left(\beta_g, \frac{1}{N_g} \left(\sum_{i \in g} X_i^2\right)^{-1} B \left(\sum_{i \in g} X_i^2\right)^{-1}\right)$$

where B_g is:

$$B_g = \text{Var} (X_{ig} u_{ig})$$

If I estimate β across G clusters I get a vector of G $\hat{\beta}_g$, and for each of these I can consistently estimate an asymptotic variance using White variance estimator since observation inside a cluster are i.i.d. Since a linear combination of normal is itself normal I can average out those variances and get an estimate of the Avar of $\hat{\beta}_{\text{ols}}$.

To test this let us simulate a dataset, first we draw a random vector of betas from a normal distribution with mean 2 and variance 4.

```
include("../src/ClusterSimulation.jl")
G = 50 #Number of clusters
Ng = 1000 # Number of individuals per cluster

β = rand(Normal(2,4),G)
```

```
50-element Vector{Float64}:
 1.500294904531966
 5.457393048728963
 6.88889006961412
 4.880289094923576
 4.4620342755462055
-2.7442893414552003
-1.07894267472046
 3.632997015579926
 5.096329143303732
-0.2133449075050513
 3.918597937699796
 1.2155488199278506
-1.8144046491502506
 ⋮
 9.827642434665236
 5.69233996308661
 9.89125116402271
 4.3636964067413135
 5.020965426261913
 1.781029873189324
-4.847079442110083
 1.9336229141951582
 3.2504765012988037
 8.010562211934872
```

```
-2.367672727706122  
5.6268667150529925
```

Then we use each of these betas to simulate a cluster according to the model in Equation 1

```
pop = hePopulation(G,β,randn(G))
```

Population with 50 clusters and with heterogeneous effect

This line creates a heterogeneous-effects population.

- **G** is the number of clusters.

- **β** is the vector of betas.

- **randn(G)** produces the vector $c_g \sim \mathcal{N}(0, 1)$.

The constructor returns a population struct that stores the cluster-level parameters used later to simulate data.

```
sam = sample(pop, Ng)
```

Cluster sample struct

This line generates a sample from the population.

- **Ng** is the number of observations per cluster.

The function draws data for all **G** clusters, each containing **Ng** observations, and returns a sample struct built according to the population's parameters.

The sample struct has a field **allclusters** that is an array of cluster structs, so we can define a function that compute $\hat{\beta}_g$ in each cluster and then broadcast it to all clusters in our sample, getting **G** estimates.

```
betahat = permutedims(stack(bols.(sam.allclusters)))[:,2]
```

```
50-element Vector{Float64}:  
 1.4486023872434155  
 5.496420622461407  
 6.871914009027232  
 4.904480013981978  
 4.456757156883328  
 -2.73891252527359  
 -1.0823851631862527  
 3.5973883687899613  
 5.125045870784402  
 -0.25728811247032385  
 3.9289564493970652  
 1.2099299249807223
```



```

-1.7858821679226795
:
9.882393245152416
5.675005139387323
9.896449203262886
4.360394542261762
5.059908583601023
1.7892183764300182
-4.830233901620227
1.8810572232973513
3.190605933669128
7.985545765646655
-2.3589695946425793
5.6233423439320775

```

And similarly define a function that compute the White Variance estimator and broadcast it to all clusters.

```
WhiteAvar.(sam.allclusters)
```

```

50-element Vector{Matrix{Float64}}:
 [0.0009409962821812022 -3.9576324747080506e-5; -3.957632474708048e-5
 0.0009034771206889339]
 [0.0009633581717642887 7.209736439499289e-6; 7.2097364394992585e-6
 0.0009300136233389295]
 [0.0010309896672866907 -2.273274606335735e-5; -2.273274606335734e-5
 0.0009422073466611694]
 [0.0009782373549542705 -2.550244761506316e-5; -2.5502447615063143e-5
 0.0008553142433319888]
 [0.0010995413680602946 -0.00011831825932905076; -0.00011831825932905075
 0.0011126774289737414]
 [0.0010832332336453148 3.3516063981807234e-5; 3.3516063981807214e-5
 0.0010033775338801566]
 [0.0010074876825842468 5.331574889363771e-5; 5.3315748893637716e-5
 0.000976915338852802]
 [0.0010819434410563728 5.466591194213505e-5; 5.4665911942135047e-5
 0.0010622734138829393]
 [0.0010231576097864347 -4.062007076754015e-5; -4.0620070767540144e-5
 0.0010728552265630777]
 [0.0010150712356153879 2.885795887665403e-5; 2.8857958876654007e-5
 0.0010302087420488754]
 [0.0009470432725958157 -3.2207777236368465e-5; -3.220777723636851e-5
 0.0010085045024832177]
 [0.0009558698691086126 5.1298173165017274e-5; 5.1298173165017254e-5
 0.0009932214363537865]
 [0.0010176672394903933 1.955295370811584e-6; 1.955295370811575e-6
 0.0010416043497960628]

```

```

:
[0.0010024636332506805 1.6582056340469407e-5; 1.6582056340469394e-5
0.0010303918784561061]
[0.0009342610886415898 -4.673012603938307e-6; -4.673012603938266e-6
0.001065852831257236]
[0.0009186653422250941 -5.0004785820420404e-5; -5.00047858204204e-5
0.0009012460738134369]
[0.0009644954203304629 -6.503122062107715e-5; -6.503122062107715e-5
0.0009901604367822201]
[0.0009500638388471534 4.5817447250222506e-5; 4.5817447250222485e-5
0.0009000909740318418]
[0.001197865283015772 -8.635700489541692e-6; -8.635700489541698e-6
0.0010742654748244975]
[0.001021912637304921 -1.3761389557657313e-6; -1.3761389557656405e-6
0.0009354338286228558]
[0.0010351050337203595 3.3429924763860474e-5; 3.342992476386056e-5
0.0009183510193008142]
[0.0009603185768346589 -2.603903701994021e-5; -2.60390370199402e-5
0.0010132405243303724]
[0.0009402042421190014 2.6901915800659203e-5; 2.690191580065921e-5
0.0010679286037735527]
[0.0009296444282159721 3.7314845414832746e-5; 3.731484541483273e-5
0.0009017144183808189]
[0.0009166617435029825 -1.7257363498241795e-5; -1.7257363498241782e-5
0.0009000354341177598]

```

We are interested in the average across clusters of β_g , the average partial effect. Note that since we have a finite amount of clusters, each of them with the same size, the APE would be:

$$\beta_{APE} \frac{1}{G} \sum_g \beta_g$$

Note that doing $\frac{1}{G} \sum_g \hat{\beta}_g$ is the same as computing $\hat{\beta}_{ols}$ in the whole sample. We define a function to do so:

```

function bols(s::clSample)
    X = getallX(s)
    Y = vcat(getallY(s)...)
    betahat = inv(X' * X) * (X'Y)
end

```

bols (generic function with 2 methods)

5 Consistency as $N_g \xrightarrow{p} \infty$

We want to test whether our estimate of the average partial effect gets better if the number of people that we draw in each cluster increases. In order to do so we ran a series of monte-carlo simulation increasing N_g .

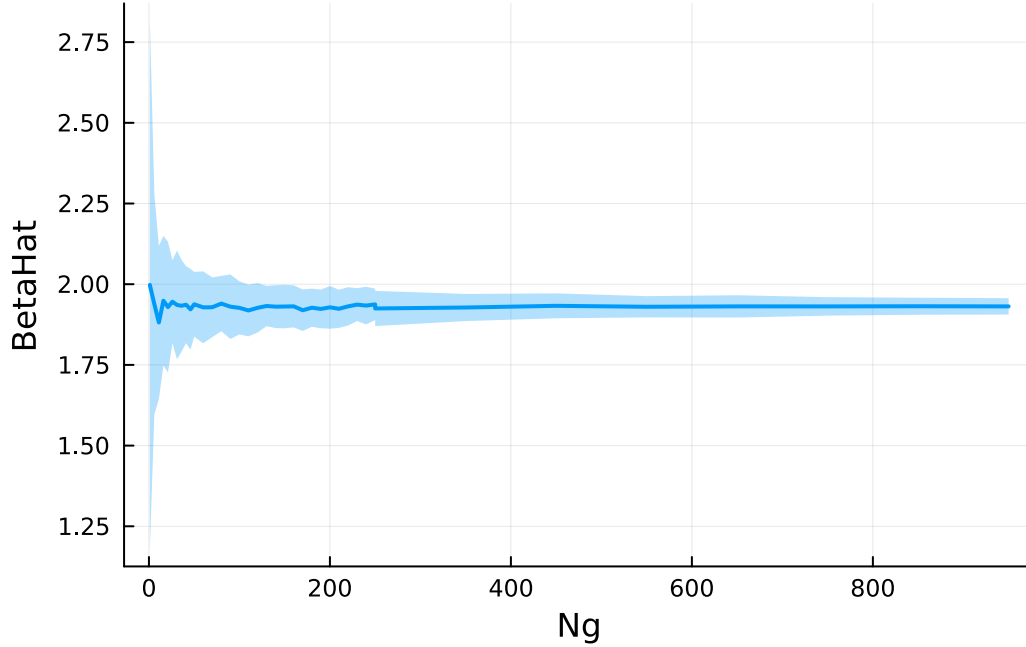


Figure 1: Consistency of betahat as N_g

What we see in Figure 1 is the result of the monte-carlo simulation, the line is the average estimates for each N_g , while the shaded area is the standard deviation, we can clearly see from the graph that as the number of observations inside a cluster grows $\hat{\beta}_{ols}^p \rightarrow \beta_{APE}$.

Bibliography

- [1] A. C. Cameron and D. L. Miller, “A Practitioner's Guide to Cluster-Robust Inference,” *Journal of Human Resources*, vol. 50, no. 2, pp. 317–372, Mar. 2015, doi: 10.3368/jhr.50.2.317.
- [2] A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge, “When Should You Adjust Standard Errors for Clustering?,” *The Quarterly Journal of Economics*, vol. 138, no. 1, pp. 1–35, Feb. 2023, doi: 10.1093/qje/qjac038.
- [3] A. C. Cameron, D. L. Miller, and others, “Robust inference with clustered data,” *Handbook of empirical economics and finance*, vol. 106, pp. 1–28, 2010.